

# Assignment III

Machine Learning

February 8, 2026

## Abstract

These exercises aim to review the Neural Network (NN) and Support Vector Machine (SVM) models for classification and regression scenarios. Please upload your solutions to Classroom before february 17th.

## Problem 1

Read section 5 from [1] to understand how partial derivatives of NNs with respect to weights and bias are computed utilizing backpropagation. In particular, sklearn uses the latter during the training phase. Write down the advantages and disadvantages of using backpropagation to compute gradients compared to finite differences.

## Problem 2

Consider the dataset from [2]. Make a Python script to solve the regression problem of forecasting city and highway fuel consumption using sklearn. Use just one NN model for the two fuel consumption categories. Write down all the assumptions, hyperparameters, and data preprocessing operations you make.

## Problem 3

Consider the following function:

$$f(x) = 1 - 2^{\sin(-x^2)}, \quad x \in \mathcal{I} = [-\pi, \pi].$$

Make a Python script to solve the regression problem associated with a NN using sklearn. Write down all the assumptions you make. Also, write down the hyperparameters you choose and explain why you picked them that way.

## Problem 4

Consider the dataset from [3]. Make a script using sklearn to solve the binary classification problem associated with classifying whether a given horse will make it or not. Use a NN model. Write down all the assumptions and data preprocessing operations you make. Incorporate the F1 score to explain your findings.

## Problem 5

Consider the dataset from [4]. Make a script using sklearn to solve the multiclass classification problem associated with classifying beers by style. Use a NN model. Write down all the assumptions and data preprocessing operations you make. Then, incorporate a confusion matrix into your analysis.

## Problem 6

Consider the general quadratic equation

$$ax^2 + bx + c = 0.$$

Do your best to classify the roots of the equation above in terms of its coefficients using a NN model with sklearn. For simplicity, you must fix the domain for  $x$  and propose a suitable domain for the coefficients. Write down all your assumptions and show off your findings using some plots.

## Problems 7

Consider problem 3 with a NN made with only one hidden layer with three neurons, logistic activation functions for such neurons, and a linear activation function in the output layer neuron. Let  $x$  be some random point in  $[-\pi, \pi]$ . Using a script in Python, compute and compare the gradient of the NN at  $x$  using central finite differences and backpropagation. You will have to make use of the formulas from problem 1. Write down your findings and assumptions.

## Problems 8 to 12

Repeat problems 2 to 6 using the SVM model from sklearn. Write down the hyperparameters you choose and all your assumptions.

## Problem 13

Consider a classification model. Why is the vector of parameters  $\theta$  orthogonal to the decision boundary? Write down your answer.

## References

- [1] C.F. Higham, D.J. Higham, **Deep Learning: An Introduction for Applied Mathematicians**, SIAM, 2019.
- [2] <https://www.kaggle.com/datasets/rinichristy/2022-fuel-consumption-ratings>
- [3] <https://www.kaggle.com/datasets/yasserh/horsesurvivalprognostication>
- [4] <https://www.kaggle.com/datasets/ankurnapa/ankurs-beer-data-set>